

Owen Henderschedt

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Employment

Postdoctoral Research Associate

IOWA STATE UNIVERSITY

Aug 2026–May 2028

Ames, IA

- Mentors: Bernard Lidický and Shira Zerbib

Education

PhD in Mathematics

AUBURN UNIVERSITY

Aug 2020–Aug 2026

Auburn, AL

- PhD Advisor: Dr. Jessica McDonald

Bachelor of Science in Actuarial Mathematics

BRYANT UNIVERSITY

Aug 2016–May 2020

Smithfield, RI

Research Interests

My research interests lie in discrete mathematics, with an emphasis on graph theory and combinatorial geometry, particularly in problems related to graph coloring, extremal graph theory and Ramsey theory.

Publications

[Journal], [Book], and [arXiv] links are clickable.

1. H. Echeverría, O. Henderschedt, *Yes, $(2K_2, K_4)$ -free graphs are recolorable.* **Submitted.** [arXiv]
2. H. Echeverría, O. Henderschedt, *On the recolorability of $(2K_2, K_4)$ -free graphs.* **Submitted.** [arXiv]
3. J. DeLeo, O. Henderschedt, C. Wells, *A finite victory over de Bruijn–Erdős in interval discrepancy.* **Submitted.** [arXiv]
4. O. Henderschedt, J. McDonald, *Extending total colorings in planar graphs.* **Submitted.** [arXiv]
5. N. Crawford, E. Heath, O. Henderschedt, C. Schwieder, S. Zerbib, *Odd Ramsey numbers of complete multipartite graphs and hypergraphs.* **Submitted.** [arXiv]
6. O. Henderschedt, J. McDonald, S. Shan, *Total coloring graphs with large minimum degree.* **Submitted.** [arXiv]
7. K. Chen, J. DeLeo, O. Henderschedt, *Graphs generated from minimal sets of finite point-set topologies.* **Submitted.** [arXiv]
8. A. Bezdek, O. Henderschedt, *Shrinking the Jung radius: Maximizing partial coverage of finite point sets.* **Submitted.** [arXiv]
9. A. Bezdek, O. Henderschedt, *Short path and short chain problems in the plane.* **New Probes into Discrete and Convex Geometry**, Bolyai Society Mathematical Studies, vol. 33, pp. 47–66, Springer, 2026. [Book]
10. O. Henderschedt, J. McDonald, *On orientations with forbidden out-degrees.* **Discrete Applied Mathematics**, 387, pp. 116–123, 2025. [Journal] [arXiv]
11. J. Anderson, H. Chau, E. Cho, N. Crawford, S. Hartke, E. Heath, O. Henderschedt, H. Kwon, Z. Zhang, *The forb-flex method for odd coloring and proper conflict-free coloring of planar graphs.* **Discrete Mathematics**, 348(12):114648, 2025. [Journal] [arXiv]

12. A. Bezdek, H. Gilroy, O. Henderschedt, A. Lakhani, *On Conway's Brussels Sprouts*. **Studia Scientiarum Mathematicarum Hungarica**, 60(1), pp. 76–90, 2023. [Journal]

Honors and Awards

- Don and Sandy Logan Endowment Research Fellowship (\$3000), Auburn University, 2025
Awarded for outstanding academic performance and research achievement at the departmental level.
- Dr. J. Earl Perry Endowed Summer Graduate Fellowship (\$2000), Auburn University, 2025
Awarded for demonstrating exceptional research potential for summer 2025 projects
- College of Sciences and Mathematics Outstanding GTA (\$500), Auburn University, 2025
Recognized at the college level as one of up to three recipients for excellence in teaching as a Graduate Teaching Assistant.
- Dr. Dean Hoffman Endowed Graduate Award (\$1200), Auburn University, 2024
Awarded for research excellence in discrete mathematics, supporting collaboration and research.
- Don and Sandy Logan Endowment Research Fellowship (\$3000), Auburn University, 2022
Awarded for outstanding academic performance and research achievement at the departmental level.
- Department of Mathematics Excellence in Research Award, Auburn Mathematics, 2022

Research Talks

Invited Talks

- Current Trends in Combinatorics, AMS Meeting #1221, Duluth, MN (October 2026)
Recoloring $(2K_2, K_4)$ -free graphs
- Recent Trends in Graph Theory, AMS Meeting #1215, Boston, MA (March 2026)
Extending total colorings in planar graphs
- Combinatorics Seminar, Georgia State University, Atlanta, GA (March 2026)
Extending total colorings in planar graphs
- Nonconventional Methods in Combinatorics, Joint Mathematics Meetings, Washington, D.C. (January 2026)
Extending total colorings in planar graphs
- Special Session on Recent Advances in Graph Theory, AMS Meeting #1210, Tulane University, New Orleans, LA (October 2025)
Extending total colorings in planar graphs
- Discrete Math and Combinatorics Seminar, University of South Carolina, virtual (September 2025)
Odd Ramsey numbers of multipartite graphs and hypergraphs
- Special Session on Sparse Graphs: Colorings, Randomness, and Ramsey Theory, Joint Mathematics Meetings, Seattle, WA (January 2025)
On orientations with forbidden out-degrees
- Special Session on Research from the Graduate Research Workshop in Combinatorics, Joint Mathematics Meetings, Seattle, WA (January 2025)
On odd colorings and proper conflict-free colorings of planar graphs with girth conditions
- Department of Mathematics and Statistics Seminar, McNeese University, virtual (October 2024)
On the number of points a given circle can cover from a finite point-set
- Graph Theory Seminar, Georgia Institute of Technology, Atlanta, GA (September 2024)
On orientations with forbidden out-degrees

- Discrete Geometry Days, Mathematics Institute of Budapest University of Technology, Budapest, HU (July 2024)
On the number of points a given circle can cover from a finite point-set
- Special Session on Discrete Geometry, AMS Meeting #1190, Mobile, AL (October 2023)
Evading squares without a map — a new approach

Department Talks

- Discrete Math Seminar, Iowa State University, Ames, IA (October 2026)
A finite victory over de Bruijn - Erdős in interval discrepancy
- The History of Mathematics for Undergraduates, Auburn University, Auburn, AL (April 2026)
On Conway's Pegboard Problem
- Combinatorics Seminar, Auburn University, Auburn, AL (October 2025)
Orientations and total coloring extensions
- REU Program in Combinatorics Seminar, Auburn University, Auburn, AL (July 2025)
Odd Ramsey numbers and a hypergraph matching tool for coloring
- Combinatorics Seminar, Auburn University, Auburn, AL (September 2024)
On orientations with forbidden out-degrees
- REU Program in Combinatorics Seminar, Auburn University, Auburn, AL (July 2024)
On orientations with forbidden out-degrees
- Graduate Student Research Seminar, Auburn University, Auburn, AL (October 2023)
Introduction to the discharging method through the lens of graph coloring variants
- REU Program in Combinatorics Seminar, Auburn University, Auburn, AL (July 2023)
The interval function and a combinatorial game on connected graphs
- Graduate Student Research Seminar, Auburn University, Auburn, AL (September 2022)
On convexity in connected graphs
- Math Club, Auburn University, Auburn, AL (October 2021)
On Conway's Brussels Sprouts — a colored combinatorial game

Teaching Experience

Iowa State University

Recitation Leader

- Fall 2026 — Calculus III (6 sections)

Auburn University

Instructor of Record

- Spring 2026 — Undergraduate Directed Reading: Introduction to Graph Coloring
- Spring 2026 — Actuarial Seminar in the Mathematics of Finance (1 section)
- Fall 2025 — Mathematics of Interest Theory (1 section)
- Fall 2024 — College Algebra (4 sections)
- Spring 2024 — Business Calculus II (1 section)
- Summer 2024 — Calculus III (1 section)
- Spring 2023 — College Algebra (3 sections)

- Summer 2023 — Calculus III (1 section)
- Summer 2022 — Calculus II (1 section)
- Fall 2021 — Pre-Calculus Trigonometry (1 section)

Recitation Leader

- Fall 2025 — Calculus I (1 section)
- Fall 2023 — Calculus I (2 sections)
- Fall 2022 — Calculus I (2 sections)

Grader and Tutor

- Spring 2022 — Pre-Calculus (2 sections)
- Spring 2021 — Tutoring Center
- Fall 2020 — Tutoring Center

Bryant University

Academic Center for Excellence: Tutor

- Fall 2017–Spring 2020 — Calculus I–III, Statistics, Interest Theory, Probability

Workshops and Summer Schools ---

- Graduate Research Workshop in Combinatorics, Iowa State University, Ames, IA (July 2026)
- Cross Community Collaborations in Combinatorics Workshop, Banff International Research Station (BIRS), Banff, AB, Canada (June 2026)
- Flag Algebra Workshop, Colorado State University, Mountain Campus, CO (June 2026)
- Preparing Future Faculty, Biggio Center, Auburn University, Auburn, AL (Fall 2025)
- Topological Methods in Combinatorics Summer School, Iowa State University, Ames, IA (May 2025)
- Discrete Geometry Days, Mathematics Institute of Budapest University of Technology, Budapest, HU (July 2024)
- Graduate Research Workshop in Combinatorics, University of Wisconsin–Milwaukee, Milwaukee, WI (May 2024)
- Workshop in Extremal Combinatorics, University of California, Anza-Borrego, CA (January 2024)
- Discrete Geometry Summer School, Rényi Institute, Budapest, HU (August 2023)
- Graduate Research Workshop in Combinatorics, University of Wyoming, Laramie, WY (July 2023)

Outreach and Service ---

- Refereed for *Discrete Mathematics* and *Discrete Applied Mathematics*
- Teaching Mentoring Program, Students mentored: 2 (2024–2025)
- Graduate Student Teaching Seminar, Talk (September 2025)
Teaching College Algebra
- Auburn University REU, Professional Development Panelist (July 2025)
- College of Science and Mathematics Destination STEM (October 2025)
Created exhibit on Ramsey theory and edge-coloring for the math carnival

Actuarial Experience

Actuarial Intern

- Actuarial Intern, Nassau Life Insurance, Hartford, CT Summer 2018, Summer 2017

Actuarial Exams

- Financial Mathematics (FM) Exam — passed April 2018
- Probability (P) Exam — passed September 2017

References

Dr. Jessica McDonald

Professor, Auburn University

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Phone: 334-844-6577

Relationship: PhD Advisor

Dr. Shira Zerbib

Associate Professor and Scott Hanna Faculty Fellow, Iowa State University

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Relationship: Research Collaborator

Dr. Andras Bezdek

Professor, Auburn University

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Relationship: Professor and Research Collaborator

Dr. Lora Merchant

Senior Lecturer and Academic Instruction Coordinator, Auburn University

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Relationship: Teaching Supervisor

Dr. Songling Shan

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Relationship: Professor and Research Collaborator